

SMART INDIA HACKATHON 2025



- Problem Statement ID – SIH25018
- Problem Statement Title- Telemedicine Access for Rural Healthcare in Nabha
- Theme- MedTech / BioTech / HealthTech
- PS Category- Software
- Team ID-
- Team Name- Bungo

CareMedha Triage Pharmacy Consult

Log in

Sign up

Describe your symptoms

Severe headache and 100 fever

Check symptoms

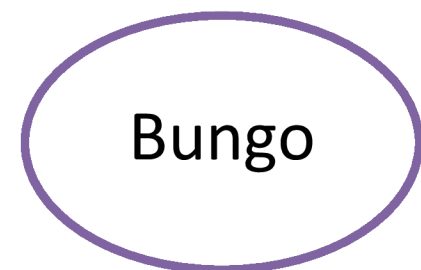
Result: Consultation

User reports a severe headache and a fever of 100°F.

ਤੁਹਾਨੂੰ ਡਾਕਟਰ ਦੀ ਸਲਾਹ ਲੈਣੀ ਚਾਹੀਦੀ ਹੈ। ਕਿਰਪਾ ਕਰਕੇ ਐਪ ਰਾਹੀਂ ਡਾਕਟਰ ਨਾਲ ਸੰਪਰਕ ਕਰੋ ਜਾਂ ਨਜ਼ਦੀਕੀ ਕਲੀਨਿਕ 'ਤੇ ਜਾਓ।

Find pharmacy

Consult a doctor



“End-to-end chain from triage → doctor → pharmacy → emergency, designed for low-connectivity & rural contexts”

How it Addresses the Problem:

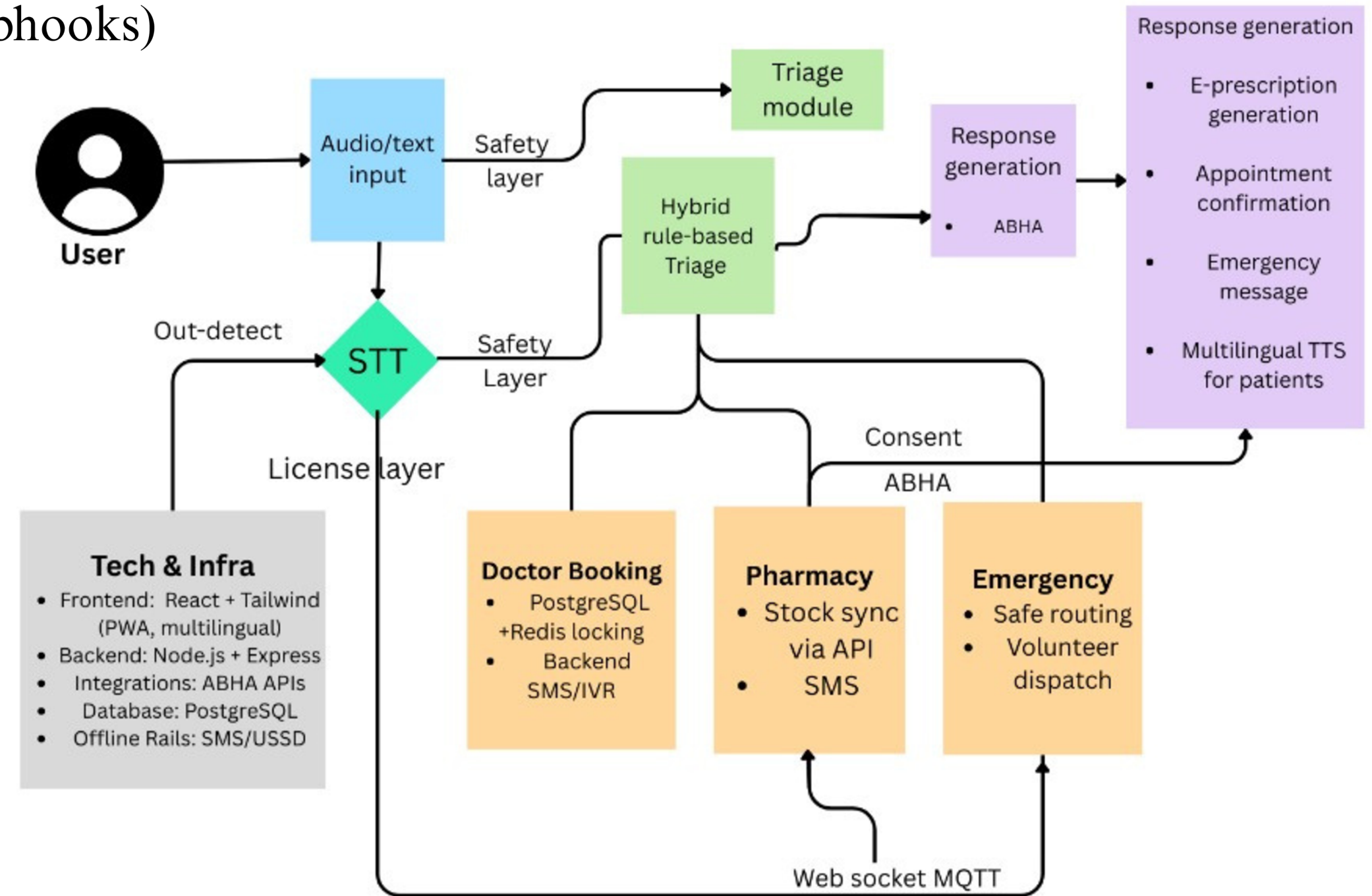
- **AI Triage (not diagnosis):** Guides patients on next steps (self-care, right doctor, or emergency), localized to Punjabi/rural context.
- **Smart Doctor Booking:** Specialty + priority scores based slot booking system; AI auto-routing ensuring scarce specialists see the right cases.
- **Telecommunication:** Video calls → auto fallback to voice call on poor bandwidth
- **Pharmacy Network:** Real-time medicine stock + fraud-safe e-prescriptions sent to patient & pharmacy simultaneously.
- **Emergency Routing:** Safe-route scores with volunteer/108 integration; geological heat map and guidance
- **Offline-first + ABHA:** Works on 2G/SMS/IVR, with multilingual voice/text and health record portability via ABHA ID

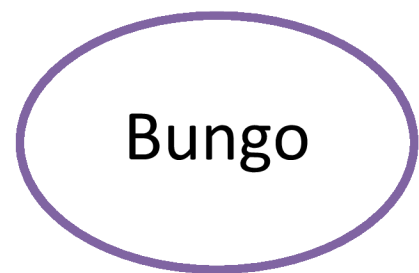
Innovation & Uniqueness:

- **Community-led adoption:** ASHA assisted.
- **Offline-first, multilingual (language toggle).**
- **Geo-adaptive AI:** Detects outbreak clusters, adapts responses.
- **Fraud-safe closed loop e-Rx for trust.**
- **Full chain coverage:** Unlike eSanjeevani/Practo/Apollo, we integrate triage, booking, pharmacy, & emergency in one platform.

- **Frontend:** React + TailwindCSS (PWA, multilingual, mobile-first)
- **Teleconsultation:** WebRTC with auto-fallback → video → voice → chat (TURN/STUN)
- **Backend:** Node.js + Express (REST APIs, webhooks)
- **Database:** PostgreSQL (patients, triage, slots, e-Rx, pharmacies, audits, ABHA map)
- **Offline Rails:** SMS/IVR gateways; store-and-forward sync
- **AI Layer:** Rules + small LLM (severity & specialty scoring, capacity-aware routing, geo-trend detection) with human-in-the-loop for risky/ambiguous cases
- **Security & Privacy:** Encryption in transit/at rest, RBAC, consent ledger, audit trails

[Demo Video for the prototype](#)





FEASIBILITY AND VIABILITY



Feasibility:

- **Works even with weak internet:** SMS, IVR, video → voice fallback.
- **Simple for doctors & pharmacies:** quick onboarding, easy dashboards.
- **Legally safe:** ABHA/ABDM aligned, e-consent, locked e-Rx.
- **Sustainable model:**
 - Govt funds rollout & health dashboards.
 - Small pharmacy/clinic subscriptions.
 - Optional micro-fees (₹1 triage, ₹5 booking), waived for BPL.
 - Private Doctors get honoraria or small tele-fees with weekly payouts.
 - Gov Doctors get reduced work load for the amount of hours they put in.

RISK	MITIGATION
Poor connectivity	Offline SMS/IVR, internet kiosks, solar devices
Low literacy	Voice prompts, icons, ASHA support
Busy doctors	Honoraria, CME credit, weekly payouts
Pharmacy adoption	Zero-effort onboarding, badges, compliance nudges
Fraud/misuse	Verified RMPs, locked e-Rx, audits
AI/data bias	Rural datasets, human oversight, audits

Impact on target audience

- **Patients & Families:** Faster guidance in Punjabi/Hindi, fewer costly trips, safer emergency response, continuity via ABHA, better chronic care.
- **Doctors & Hospitals:** Fewer trivial walk-ins, organized queues, richer triage summaries, legal safety, professional recognition.
- **Pharmacies:** Higher footfall, demand-driven stocking, verified e-Rx, optional delivery revenue.
- **Government & Public Health:** Real-time village-level health & stock data, decongested OPDs, equitable access, stronger disaster response.

Benefits

- **Social:** Timely care, safer emergencies, local-language empowerment.
- **Economic:** Reduced travel costs & wage loss; less hospital overload saves govt resources.
- **Public Health:** Early outbreak detection, drug stock monitoring.
- **Environmental:** Lower unnecessary travel → smaller carbon footprint.

- **Telemedicine Guidelines 2020** → consent, identity, safe prescribing norms.
- **ABDM / ABHA** → record portability & interoperability with national digital health mission.
- **eSanjeevani** → we complement the platform by feeding qualified demand and offline-first reach.
- **WHO & Rural Health Toolkits** → identified barriers like connectivity & literacy, leading to our offline-first design.
- **India CHW Chatbots** → co-designed with ASHAs, showing why community-assisted flows succeed.

